

SoftCart E-Commerce Performance Report

Warehouse-backed sales analysis generated from a validated PostgreSQL dimensional model.

POSTGRES SQL WAREHOUSE

300K FACT ROWS

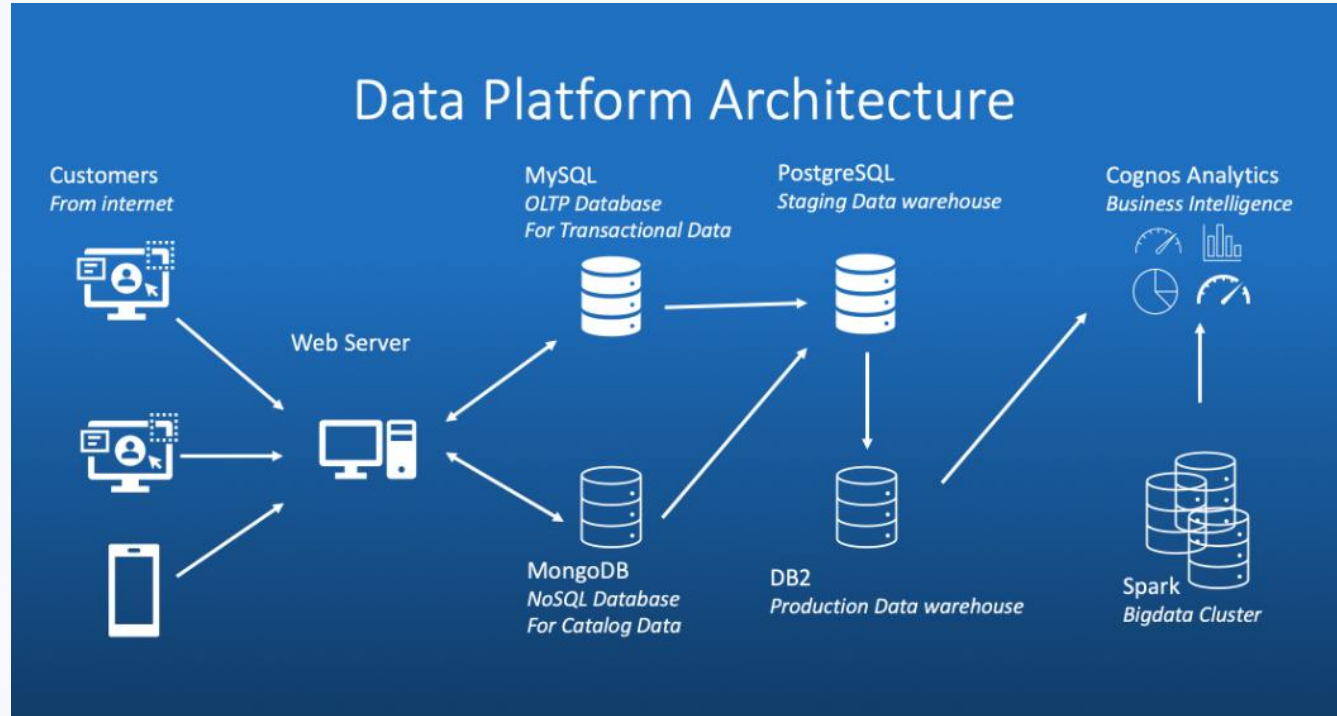
PORTFOLIO ARTEFACT

The report turns the course-derived SoftCart warehouse into an editable business-facing presentation: validated loads, reproducible SQL outputs, and concise sales insights.

Generated outputs: total_sales_per_country.csv · total_sales_per_category.csv · sales_by_year_country_rollup.csv · average_sales_cube.csv

Platform architecture

Data moves from customer-facing activity into operational stores, then into the warehouse and analytics layer.



Architecture interpretation

Web activity feeds operational data stores: MySQL for transactional sales and MongoDB for catalogue-style data.

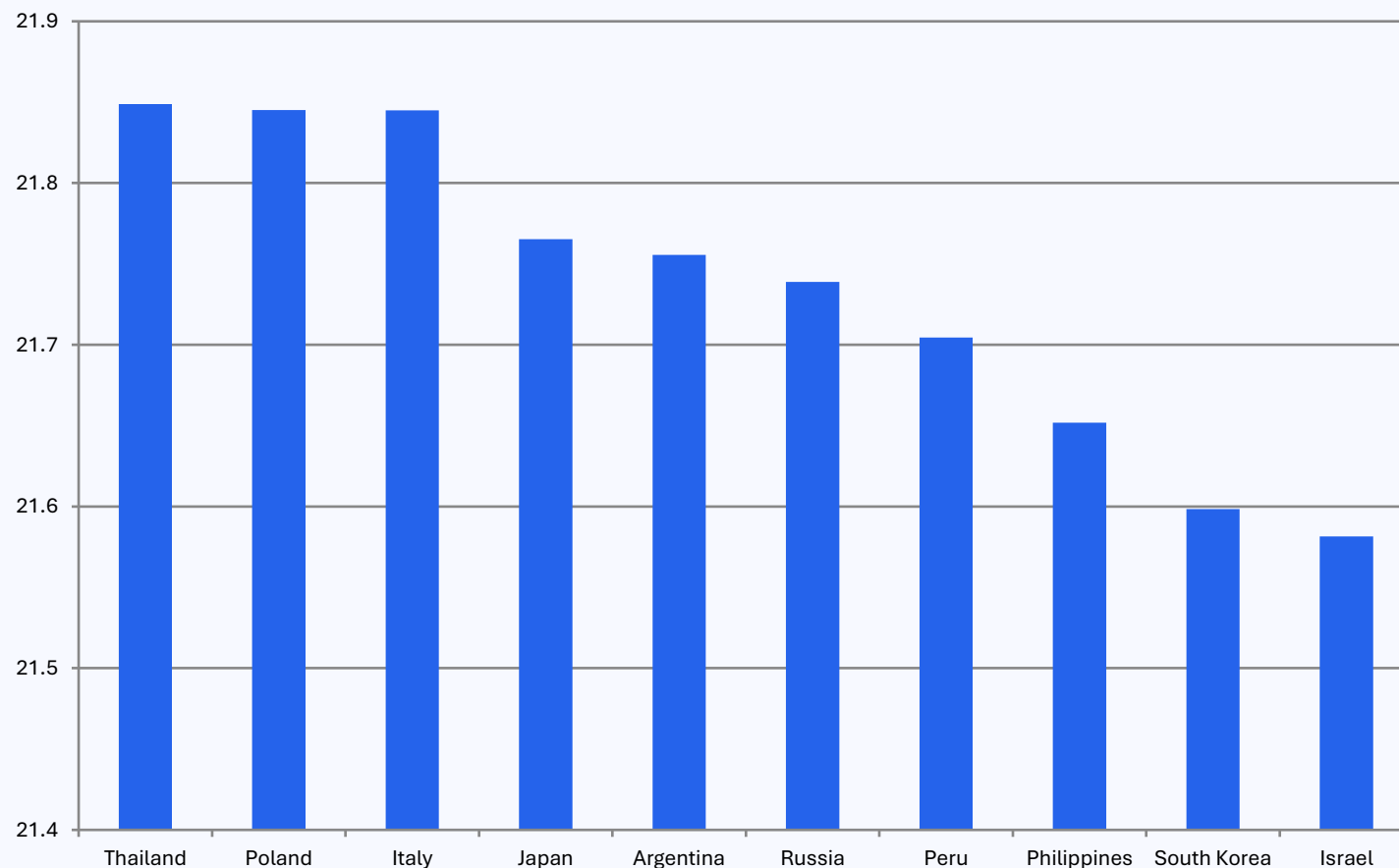
PostgreSQL acts as a staging/warehouse layer in this portfolio implementation, allowing dimensional sales analysis and CSV exports.

BI reporting and Spark analytics sit downstream: reporting for business consumption, Spark for scalable workflow practice.

Note: the original course diagram refers to PostgreSQL staging and DB2 production warehousing; this repo validates the warehouse layer locally in PostgreSQL.

Country sales are tightly clustered

Top ten countries differ by less than 1.3% from first to tenth place, suggesting broad geographic balance.



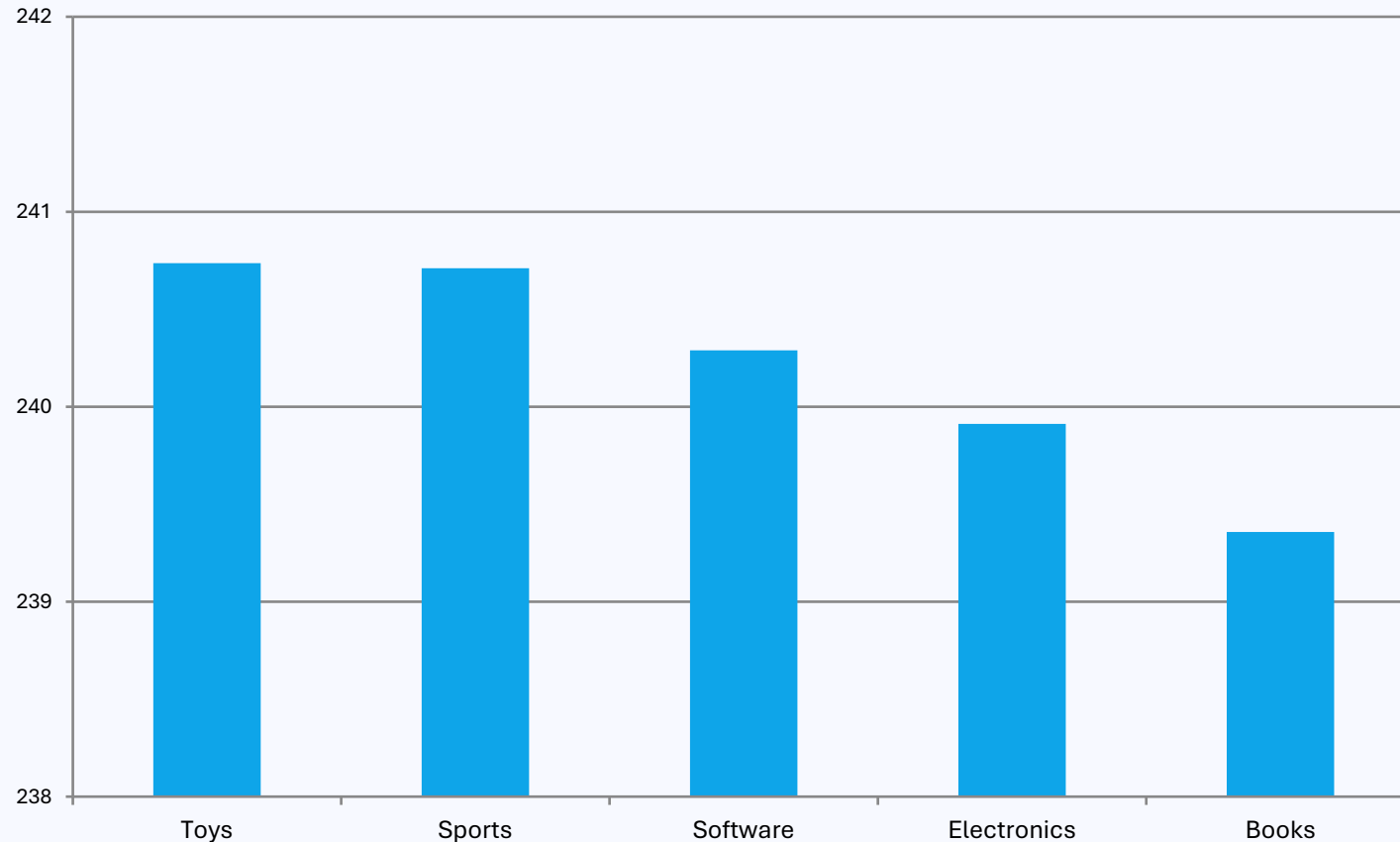
Thailand ranks first at £21.85m, but Poland and Italy are almost identical. The useful message is not “Thailand dominates”; it does not. The correct reading is geographic sales parity across top markets.

Portfolio angle: this proves the warehouse can support country-level commercial reporting from dimensional joins.

Source: analytics/total_sales_per_country.csv

Category performance is deliberately balanced

The five product categories sit around £239m–£241m, so the dataset shows little real category separation.



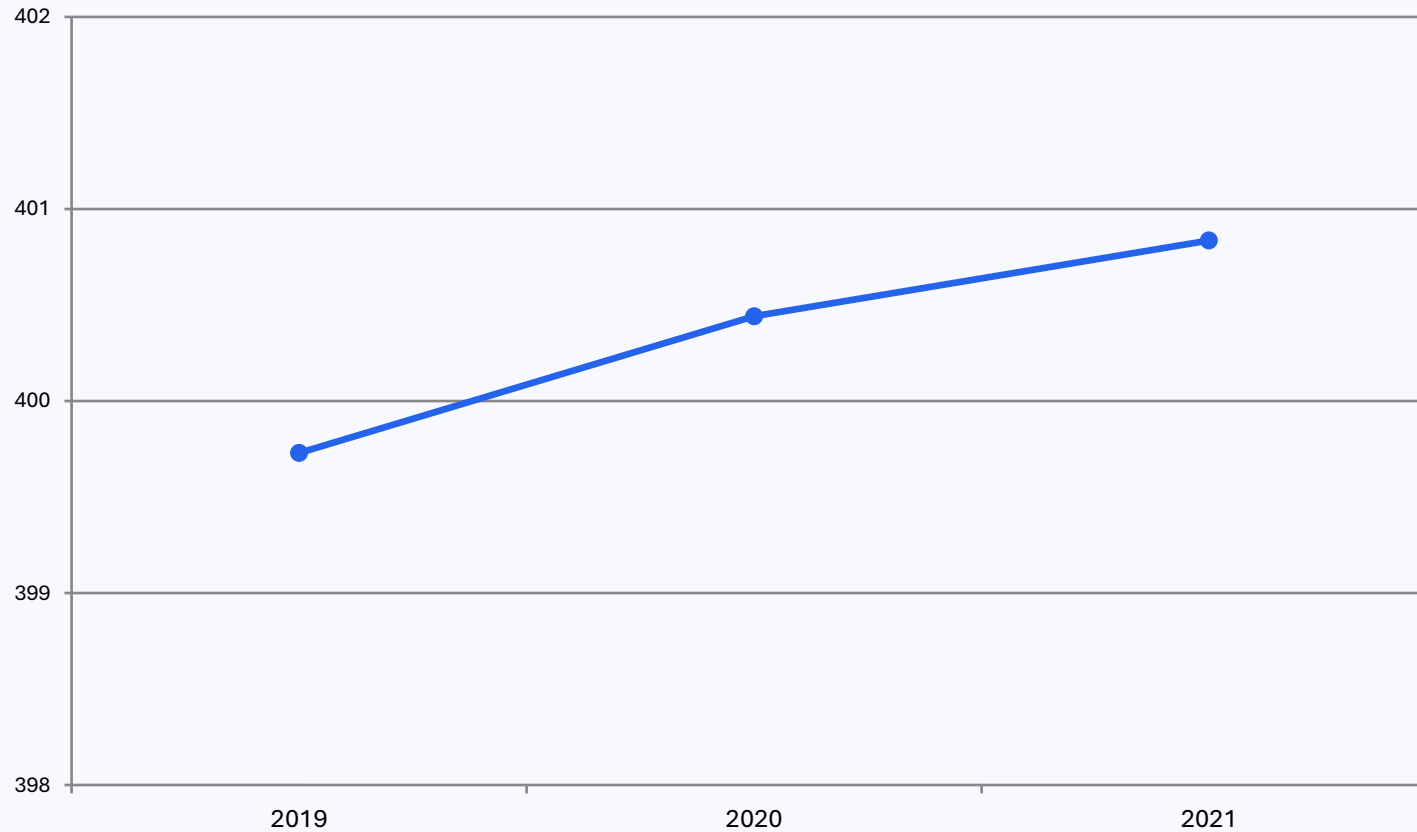
Toys is the highest category at £240.74m, followed very closely by Sports and Software. Do not oversell this as a sharp commercial insight; the safer interpretation is that the dataset is evenly distributed.

Portfolio angle: this slide demonstrates category-level aggregation from the dimensional warehouse.

Source: analytics/total_sales_per_category.csv

Yearly sales trend is stable

Sales increase slightly year-on-year, but the movement is small relative to total volume.



Total sales rise from £399.73m in 2019 to £400.84m in 2021. That is a weak growth signal, not a dramatic trend.

This is why the report should focus on reproducibility and warehouse validation rather than exaggerated business conclusions.

Portfolio angle: the year-level trend is generated by joining FactSales to DimDate and grouping by year.

Source: dimensional join between FactSales and DimDate

What the project actually proves

The value is not a flashy dashboard; it is a reproducible data engineering workflow.

Validated workflow

Raw course files were reorganised into a clear repository structure with data, SQL, warehouse, analytics, ETL, NoSQL, Spark, reports and documentation layers.

Warehouse SQL was corrected to match the actual schema and successfully loaded into PostgreSQL.

Analytical outputs were exported as reusable CSVs rather than being left as screenshots.

Evidence generated

300,000

FactSales rows loaded

1,096

date dimension records

56

countries analysed in dimensional model

£1.20B

total recorded sales amount

Next upgrade

Add Docker Compose and a one-command run script. That would move the project from “clean portfolio artefact” to “proper reproducible engineering demo”.

Repository: github.com/barbavegeta/softcart-data-engineering-platform